

R Bootcamp Day 2

The tidyverse

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Today

Today

Goals

- Familiarity with the core of the tidyverse for data manipulation
- Ability to use `tibble`, `dplyr`, and `purrr` over base R equivalents
- Practice with actual datasets in R
- Tomorrow: `ggplot2`

Sources

- Wickham, Çetinkaya-Rundel, and Grolemund, *R for Data Science*, 2e
 - [Ch. 3: Data transformation](<https://r4ds.hadley.nz/data-transform.html>) (especially §§3.1-3.5)
- Harvard IQSS, *Math Prefresher for Political Scientists*
 - [Ch. 9: Objects, Functions, Loops](https://iqss.github.io/prefresher/13_functions_obj_loops.html)

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 - [Ch. 19: Joins](<https://r4ds.hadley.nz/joins.html>) (especially §§19.2-19.3)
- Harvard IQSS, *Math Prefresher for Political Scientists*
 - [Ch. 9: Objects, Functions, Loops](https://iqss.github.io/prefresher/13_functions_obj_loops.html)
 - [Ch. 11: Joins and Merges](https://iqss.github.io/prefresher/15_project-dempeace.html)

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 - [Ch. 3: Data transformation](<https://r4ds.hadley.nz/data-transform.html>) (especially §§3.1-3.5)
 - [Ch. 19: Joins](<https://r4ds.hadley.nz/joins.html>) (especially §§19.2-19.3)
 - [Ch. 26: Iteration](<https://r4ds.hadley.nz/iteration.html>) (especially §§26.2-26.3)
- Harvard IQSS, *Math Prefresher for Political Scientists*
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The tidyverse

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 - each observation is a row and each column is a variable

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The tidyverse

- A collection of R packages for data science
- Packages designed around tidy data
 - each observation is a row and each column is a variable
- Core packages today: tibble, dplyr, tidyr, and purrr

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2     4.0.3      ✓ tibble     3.3.1
✓ lubridate   1.9.5      ✓ tidyr      1.3.2
✓ purrr       1.2.2

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package to force all conflicts to become errors
```

Tibble

- A tibble is a data frame with more predictable behavior

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- Printing shows the first rows and the columns that fit

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Tibble

- A tibble is a data frame with more predictable behavior
- Printing shows the first rows and the columns that fit
- Column types appear below the column names

```
counties
```

```
# A tibble: 3,144 × 7
  GEOID state  county      income income_moe  rent rent_moe
  <chr> <chr>   <chr>      <dbl>      <dbl> <dbl>   <dbl>
1 01001 Alabama Autauga County  69841      5512  1200     75
2 01003 Alabama Baldwin County  75019      2751  1211     42
3 01005 Alabama Barbour County  44290      2762   644     41
# i 3,141 more rows
```



Figure 1: Tibble the cat

Printing a tibble

```
counties
```

```
# A tibble: 3,144 × 7
  GEOID state  county      income income_moe  rent rent_moe
  <chr> <chr>   <chr>      <dbl>     <dbl> <dbl>   <dbl>
1 01001 Alabama Autauga County  69841      5512  1200      75
2 01003 Alabama Baldwin County  75019      2751  1211      42
3 01005 Alabama Barbour County  44290      2762   644      41
# i 3,141 more rows
```

Tibble printing



Figure 2: Tibble's `print()`

Creating a tibble

- `tibble()` builds a data frame from named columns

Creating a tibble

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- Columns have equal length or length one

Creating a tibble

- `tibble()` builds a data frame from named columns
- Columns have equal length or length one

```
sample_respondents <- tibble(  
  id = c(101L, 102L, 103L),  
  party = c('Democrat', 'Independent', 'Republican'),  
  trust_government = c(3L, 5L, 2L)  
)
```

```
tibble(id = seq_len(3L), enrolled = TRUE)
```

```
# A tibble: 3 × 2  
  id enrolled  
  <int> <lgl>  
1     1 TRUE  
2     2 TRUE  
3     3 TRUE
```

Inspecting a tibble

Inspect before transforming data

- `glimpse()` displays each column and its type

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- `names()` returns column names

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- `dim()` returns rows and columns

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Inspect before transforming data

- `glimpse()` displays each column and its type
- `names()` returns column names
- `dim()` returns rows and columns

```
counties |>  
  select(state, county, income, rent) |>  
  glimpse()
```

```
Rows: 3,144  
Columns: 4  
$ state  <chr> "Alabama", "Alabama", "Alabama", "Alabama", "Alabama", "Alabama...  
$ county <chr> "Autauga County", "Baldwin County", "Barbour County", "Bibb Cou...  
$ income <dbl> 69841, 75019, 44290, 51215, 61096, 36723, 44881, 55826, 49295, ...  
$ rent   <dbl> 1200, 1211, 644, 802, 743, 635, 722, 804, 850, 750, 855, 653, 7...
```

Extracting a column

\$ and [[] return a column as a vector

Extracting a column

\$ and [[return a column as a vector

```
class(counties$income)
class(counties[['income']])
```

```
[1] "numeric"
[1] "numeric"
```

Extracting a column

\$ and [[return a column as a vector

```
class(counties$income)  
class(counties[['income']])
```

```
[1] "numeric"  
[1] "numeric"
```

Both approaches return a vector

Extracting a column

\$ and [[return a column as a vector

```
class(counties$income)  
class(counties[['income']])
```

```
[1] "numeric"  
[1] "numeric"
```

Both approaches return a vector

[return returns a one-column tibble

```
class(counties['income'])
```

```
[1] "tbl_df"      "tbl"        "data.frame"
```

Data transformation

dplyr verbs

- Rows
 - `filter()/slice()`: subset to specific rows
 - `arrange()`: reorder by a variable

dplyr verbs

- Rows
 - `filter()/slice()`: subset to specific rows
 - `arrange()`: reorder by a variable
- Columns
 - `select()`: subset to specific columns
 - `mutate()`: create or update columns
 - `rename()`: rename specific columns

dplyr verbs

- Rows
 - `filter()/slice()`: subset to specific rows
 - `arrange()`: reorder by a variable
- Columns
 - `select()`: subset to specific columns
 - `mutate()`: create or update columns
 - `rename()`: rename specific columns
- Groups
 - `group_by()`: identify groupings in data
 - `summarize()`: reduce groups to summaries

filter()

`filter()` keeps rows that satisfy a logical condition

filter()

`filter()` keeps rows that satisfy a logical condition

```
counties |>
  filter(income ≥ 75000)
```

```
# A tibble: 716 × 7
  GEOID state  county      income income_moe  rent rent_moe
  <chr> <chr>   <chr>      <dbl>     <dbl> <dbl>   <dbl>
1 01003 Alabama Baldwin County  75019      2751  1211     42
2 01051 Alabama Elmore County   75553      3115  1014     48
3 01083 Alabama Limestone County 83534      2861   908     49
# i 713 more rows
```

Multiple conditions: Equivalent to AND

Commas require every condition to be TRUE

Multiple conditions: Equivalent to AND

Commas require every condition to be TRUE

```
counties |>
  filter(state == 'New York', income ≥ 75000)
```

```
# A tibble: 19 × 7
  GEOID state    county    income income_moe  rent rent_moe
  <chr> <chr>    <chr>    <dbl>    <dbl> <dbl>    <dbl>
1 36001 New York Albany County 83149      1788  1252      22
2 36021 New York Columbia County 83619      3965  1199      45
3 36027 New York Dutchess County 97273      2529  1522      45
# i 16 more rows
```

Multiple conditions: Equivalent to AND

Commas require every condition to be TRUE

```
counties |>
  filter(state = 'New York', income ≥ 75000)
```

```
# A tibble: 19 × 7
  GEOID state    county    income income_moe  rent rent_moe
  <chr> <chr>    <chr>    <dbl>    <dbl> <dbl>    <dbl>
1 36001 New York Albany County 83149      1788  1252      22
2 36021 New York Columbia County 83619      3965  1199      45
3 36027 New York Dutchess County 97273      2529  1522      45
# i 16 more rows
```

which is the same as:

```
counties |>
  filter(state = 'New York') |>
  filter(income ≥ 75000)
```


Missing values

```
scores <- tibble(  
  id = seq_len(4L),  
  score = c(88L, NA, 91L, 76L)  
)
```

What happens with NAs?

```
scores |> filter(score > 90)
```

Missing values

```
scores <- tibble(  
  id = seq_len(4L),  
  score = c(88L, NA, 91L, 76L)  
)
```

What happens with NAs?

```
scores |> filter(score > 90)
```

```
# A tibble: 1 × 2  
  id score  
  <int> <int>  
1     3    91
```

arrange()

- `arrange( )` orders rows by one or more columns

arrange()

- `arrange( )` orders rows by one or more columns
- Use `desc( )` for descending order

arrange()

- `arrange()` orders rows by one or more columns
- Use `desc()` for descending order

```
counties |>  
  arrange(desc(income))
```

```
# A tibble: 3,144 × 7  
  GEOID state      county      income income_moe  rent rent_moe  
  <chr> <chr>      <chr>      <dbl>      <dbl> <dbl>      <dbl>  
1 51107 Virginia Loudoun County 178707      3411  2317         39  
2 06085 California Santa Clara County 159674      1962  2814         16  
3 06081 California San Mateo County 156000      3019  2893         36  
# i 3,141 more rows
```

distinct()

`distinct()` keeps unique values

distinct()

distinct() keeps unique values

```
counties |> distinct(state)
```

```
# A tibble: 51 × 1
  state
<chr>
1 Alabama
2 Alaska
3 Arizona
# i 48 more rows
```

distinct()

distinct() keeps unique values

```
counties |> distinct(state)
```

```
# A tibble: 51 × 1  
  state  
  <chr>  
1 Alabama  
2 Alaska  
3 Arizona  
# i 48 more rows
```

Use multiple columns for unique combinations

```
counties |> distinct(state, county) |> nrow()
```

```
[1] 3144
```


select()

`select()` keeps, drops, or reorders columns:

select()

select() keeps, drops, or reorders columns:

```
counties |>  
  select(state, county, income, rent)
```

```
# A tibble: 3,144 × 4  
  state    county      income  rent  
  <chr>   <chr>      <dbl> <dbl>  
1 Alabama Autauga County  69841  1200  
2 Alabama Baldwin County  75019  1211  
3 Alabama Barbour County  44290   644  
# i 3,141 more rows
```

select(): ranges and exclusions

We can also select with ranges ...

```
counties |> select(state:income) |> names()
```

```
[1] "state" "county" "income"
```

select(): ranges and exclusions

We can also select with ranges ...

```
counties |> select(state:income) |> names()
```

```
[1] "state" "county" "income"
```

... and with exclusions:

```
counties |> select(-c(income_moe, rent_moe)) |> names()
```

```
[1] "GEOID" "state" "county" "income" "rent"
```

Selection helpers

We can select based on names using `starts_with()`, `ends_with()`, and `contains()`

```
counties |> select(starts_with('rent')) |> names()
```

```
[1] "rent"      "rent_moe"
```

Selection helpers

We can select based on names using `starts_with()`, `ends_with()`, and `contains()`

```
counties |> select(starts_with('rent')) |> names()
```

```
[1] "rent"      "rent_moe"
```

Or with a helper function inside `where()`:

```
counties |> select(where(is.numeric)) |> names()
```

```
[1] "income"      "income_moe" "rent"        "rent_moe"
```

rename()

```
counties |>
  rename(
    median_renter_income = income,
    median_gross_rent = rent
  )
```

```
# A tibble: 3,144 × 7
  GEOID state  county median_renter_income income_moe median_gross_rent rent_moe
  <chr> <chr>   <chr>           <dbl>         <dbl>           <dbl>    <dbl>
1 01001 Alaba... Autau...       69841         5512           1200      75
2 01003 Alaba... Baldw...       75019         2751           1211      42
3 01005 Alaba... Barbo...       44290         2762            644      41
# i 3,141 more rows
```

relocate()

Move columns to the front:

```
counties |> relocate(income, rent) |> names()
```

```
[1] "income"      "rent"        "GEOID"       "state"       "county"  
[6] "income_moe"  "rent_moe"
```


relocate()

Move columns to the front:

```
counties |> relocate(income, rent) |> names()
```

```
[1] "income"      "rent"        "GEOID"       "state"       "county"  
[6] "income_moe"  "rent_moe"
```

Or to a specific location:

```
counties |> relocate(county, .after = state) |> names()
```

```
[1] "GEOID"       "state"       "county"      "income"      "income_moe"  
[6] "rent"        "rent_moe"
```

mutate()

`mutate()` creates new columns or replaces old ones

mutate()

mutate() creates new columns or replaces old ones

```
counties |>
  mutate(
    income_thousands = income / 1000,
    rent_hundreds = rent / 100
  ) |>
  select(state, county, income_thousands, rent_hundreds)
```

```
# A tibble: 3,144 × 4
  state    county    income_thousands rent_hundreds
<chr>    <chr>          <dbl>          <dbl>
1 Alabama Autauga County    69.8           12
2 Alabama Baldwin County   75.0           12.1
3 Alabama Barbour County   44.3            6.44
# i 3,141 more rows
```

Keeping only specific columns

`.keep = 'none'` keeps only the updated columns

```
counties |>
  mutate(
    income_thousands = income / 1000,
    rent_hundreds = rent / 100,
    .keep = 'none'
  )
```

```
# A tibble: 3,144 × 2
  income_thousands rent_hundreds
      <dbl>          <dbl>
1         69.8         12
2         75.0        12.1
3         44.3         6.44
# i 3,141 more rows
```

Keeping only specific columns

`.keep = 'none'` keeps only the updated columns

```
counties |>
  mutate(
    income_thousands = income / 1000,
    rent_hundreds = rent / 100,
    .keep = 'none'
  )
```

```
# A tibble: 3,144 × 2
  income_thousands rent_hundreds
      <dbl>         <dbl>
1         69.8         12
2         75.0        12.1
3         44.3         6.44
# i 3,141 more rows
```

Before 2025, this was `transmute()`

Conditional variables: if_else()

```
counties |>
  mutate(
    rent_level = if_else(
      rent >= median(rent, na.rm = TRUE),
      'at or above median',
      'below median'
    )
  ) |>
  select(state, county, rent, rent_level)
```

```
# A tibble: 3,144 × 4
  state    county      rent rent_level
<chr>    <chr>    <dbl> <chr>
1 Alabama Autauga County  1200 at or above median
2 Alabama Baldwin County 1211 at or above median
3 Alabama Barbour County   644 below median
# i 3,141 more rows
```

Complex conditional variables: case_when()

```
counties |>
  mutate(
    income_group = case_when(
      is.na(income) ~ 'missing',
      income ≥ 100000 ~ 'high',
      income ≥ 75000 ~ 'middle',
      .default = 'low'
    )
  ) |>
  select(state, county, income, income_group)
```

```
# A tibble: 3,144 × 4
  state    county      income income_group
<chr>    <chr>      <dbl> <chr>
1 Alabama Autauga County  69841 low
2 Alabama Baldwin County  75019 middle
3 Alabama Barbour County  44290 low
# i 3,141 more rows
```

Exercise

- Keep counties in New York or Texas
- Create `income_thousands` from `income`
- Keep `state`, `county`, `income`, and `income_thousands`
- Order from highest to lowest income

Write this as one pipeline

A solution

```
counties |>
  filter(state %in% c('New York', 'Texas')) |>
  mutate(income_thousands = income / 1000) |>
  select(state, county, income, income_thousands) |>
  arrange(desc(income))
```

```
# A tibble: 316 × 4
  state      county      income income_thousands
  <chr>      <chr>      <dbl>          <dbl>
1 New York Nassau County  143408          143.
2 New York Suffolk County 128329          128.
3 New York Putnam County  127405          127.
# i 313 more rows
```

Pivots in tidy

Long and wide data

Wide data

Each observation is a row

```
wide <- tribble(  
  ~id, ~v1, ~v2, ~v3,  
  'A', 100L, 120L, 105L,  
  'B', 80L, 90L, 90L  
)
```

Long data

Observations are split across rows

```
long <- tribble(  
  ~id, ~statistic, ~value,  
  'A', 'v1', 100L,  
  'A', 'v2', 120L,  
  'A', 'v3', 105L,  
  'B', 'v1', 80L,  
  'B', 'v2', 90L,  
  'B', 'v3', 90L,  
)
```

pivot_wider()

`pivot_wider()` shortens the data by adding columns

```
long |>
  pivot_wider(
    names_from = statistic,
    values_from = value
  )
```

```
# A tibble: 2 × 4
  id      v1    v2    v3
<chr> <int> <int> <int>
1 A      100   120   105
2 B       80    90    90
```

`pivot_longer()`

`pivot_longer()` lengthens the data by turning columns into rows

```
wide |>
  pivot_longer(
    cols = v1:v3,
    names_to = 'statistic',
    values_to = 'value'
  )
```

```
# A tibble: 6 × 3
  id      statistic value
<chr> <chr>      <int>
1 A      v1         100
2 A      v2         120
3 A      v3         105
# i 3 more rows
```

Other weird data shapes

- Most can be **tidied** with tidyr

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- Nested data? Some variables are lists
 - `unnest()`, `unnest_wider()`, `unnest_longer()`

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- Nested data? Some variables are lists
 - `unnest()`, `unnest_wider()`, `unnest_longer()`
- Data should be nested?
 - Are you sure?
 - `nest()`

Other weird data shapes

- Most can be **tidied** with tidyr
- Nested data? Some variables are lists
 - `unnest()`, `unnest_wider()`, `unnest_longer()`
- Data should be nested?
 - Are you sure?
 - `nest()`
- Handle NA values
 - `drop_na()` or `replace_na()`
 - Missing within groups: `fill()`

Other weird data shapes

- Most can be **tidied** with tidyr
- Nested data? Some variables are lists
 - `unnest()`, `unnest_wider()`, `unnest_longer()`
- Data should be nested?
 - Are you sure?
 - `nest()`
- Handle NA values
 - `drop_na()` or `replace_na()`
 - Missing within groups: `fill()`
- Missing combinations (e.g., all weeks should have 7 days)
 - `expand()` for new data or `complete()` to supplement

Groups and summaries

summarize()

`summarize()` reduces rows to summary values

summarize()

summarize() reduces rows to summary values

```
counties |>
  summarize(
    n = n(),
    mean_income = mean(income, na.rm = TRUE),
    median_rent = median(rent, na.rm = TRUE),
    .groups = 'drop'
  )
```

```
# A tibble: 1 × 3
      n mean_income median_rent
<int>      <dbl>      <dbl>
1  3144    66073.        854
```

summarize()

summarize() reduces rows to summary values

```
counties |>
  summarize(
    n = n(),
    mean_income = mean(income, na.rm = TRUE),
    median_rent = median(rent, na.rm = TRUE),
    .groups = 'drop'
  )
```

```
# A tibble: 1 × 3
      n mean_income median_rent
<int>      <dbl>      <dbl>
1  3144    66073.        854
```

If there are no groups, summarize() treats all rows as one group

group_by()

`group_by()` marks a tibble as having groups

group_by()

group_by() marks a tibble as having groups

```
counties |>
  group_by(state) |>
  summarize(
    n = n(),
    mean_income = mean(income, na.rm = TRUE),
    median_rent = median(rent, na.rm = TRUE)
  )
```

```
# A tibble: 51 × 4
  state      n mean_income median_rent
<chr>  <int>      <dbl>      <dbl>
1 Alabama    67    54196.         762
2 Alaska    30    79407.        1243
3 Arizona   15    62663.         983
# i 48 more rows
```


Grouped mutation

`mutate()` will respect groups for vectorized operations

Grouped mutation

`mutate()` will respect groups for vectorized operations

```
counties |>
  group_by(state) |>
  mutate(
    income_difference = income - mean(income, na.rm = TRUE)
  ) |>
  select(state, county, income, income_difference) |>
  ungroup()
```

```
# A tibble: 3,144 × 4
  state    county      income income_difference
<chr>    <chr>      <dbl>          <dbl>
1 Alabama Autauga County  69841         15645.
2 Alabama Baldwin County 75019         20823.
3 Alabama Barbour County 44290         -9906.
# i 3,141 more rows
```

count()

```
counties |> count(state)
```

```
# A tibble: 51 × 2  
  state      n  
  <chr>  <int>  
1 Alabama    67  
2 Alaska     30  
3 Arizona    15  
# i 48 more rows
```

count()

```
counties |> count(state)
```

```
# A tibble: 51 × 2  
  state      n  
  <chr>  <int>  
1 Alabama    67  
2 Alaska    30  
3 Arizona    15  
# i 48 more rows
```

count(state) groups by state and counts rows

Equivalent to:

```
counties |> group_by(state) |> summarize(n = n())
```

Multiple summaries: across()

```
counties |>
  summarize(
    across(
      c(income, rent),
      list(
        mean = \(x) mean(x, na.rm = TRUE),
        sd = \(x) sd(x, na.rm = TRUE)
      )
    )
  )
```

```
# A tibble: 1 × 4
  income_mean income_sd rent_mean rent_sd
    <dbl>      <dbl>    <dbl>    <dbl>
1   66073.    17387.     938.     302.
```

Multiple summaries: across()

```
counties |>
  summarize(
    across(
      c(income, rent),
      list(
        mean = \(x) mean(x, na.rm = TRUE),
        sd = \(x) sd(x, na.rm = TRUE)
      )
    )
  )
```

```
# A tibble: 1 × 4
  income_mean income_sd rent_mean rent_sd
    <dbl>      <dbl>    <dbl>    <dbl>
1   66073.    17387.     938.     302.
```

`across()` applies the same functions to several columns

Multiple summaries: across()

`across( )` simplifies making summaries:

Multiple summaries: across()

across() simplifies making summaries:

```
counties |>
  summarize(across(where(is.numeric), \(x) mean(x, na.rm = TRUE)))
```

```
# A tibble: 1 × 4
  income income_moe  rent rent_moe
  <dbl>      <dbl> <dbl>    <dbl>
1 66073.      5068.  938.     73.6
```


Exercise

Compare counties above and below the median income.

- Exclude counties with missing income
- Label counties as at or above median or below median
- For each group, report n and the mean income and rent using `across()`

Write this as one pipeline

A solution

```
sol <- counties |>
  filter(!is.na(income)) |>
  mutate(
    income_level = if_else(
      income >= median(income),
      'at or above median',
      'below median'
    )
  ) |>
  group_by(income_level) |>
  summarize(
    n = n(),
    across(c(income, rent), \(x) mean(x, na.rm = TRUE)),
    .groups = 'drop'
  ) |>
  arrange(desc(income))
```

A solution

```
sol
```

```
# A tibble: 2 × 4  
  income_level      n income  rent  
  <chr>      <int>  <dbl> <dbl>  
1 at or above median 1571 78668. 1090.  
2 below median      1571 53479.  787.
```

Joins

Joins

Join tables by a fixed key

- A join combines columns from related tables
- A key identifies which rows should match

Join tables by a fixed key

- A join combines columns from related tables
- A key identifies which rows should match

```
party <- tibble(  
  id = c(101L, 102L, 103L),  
  party = c('Democrat', 'Independent', 'Republican')  
)  
  
attitudes <- tibble(  
  id = c(101L, 102L, 104L),  
  trust_government = c(3L, 5L, 2L)  
)
```

Join tables by a fixed key

- A join combines columns from related tables
- A key identifies which rows should match

```
party <- tibble(  
  id = c(101L, 102L, 103L),  
  party = c('Democrat', 'Independent', 'Republican')  
)  
  
attitudes <- tibble(  
  id = c(101L, 102L, 104L),  
  trust_government = c(3L, 5L, 2L)  
)
```

A key is (often) just a numeric ID

Check the key

A key should identify the intended unit of observation

```
attitudes |>  
  count(id) |>  
  filter(n > 1L)
```

```
# A tibble: 0 × 2  
# i 2 variables: id <int>, n <int>
```


Check the key

A key should identify the intended unit of observation

```
attitudes |>  
  count(id) |>  
  filter(n > 1L)
```

```
# A tibble: 0 × 2  
# i 2 variables: id <int>, n <int>
```

An empty result means no duplicate keys were found

Left and right tables

left

id	party
101	Democrat
102	Independent
103	Republican
104	Democrat
106	Republican

right

id	trust
102	5
103	3
104	2
105	4
106	7

Types of joins

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Which rows are saved for each join?

```
left |>  
  *_join(by = 'id')
```

Joining real data

counties and county_pop use different names for the same key

```
counties_joined <- counties |>
  left_join(county_pop, join_by(GEOID = fips))

counties_joined |>
  select(GEOID, state, county, pop, region, division)
```

```
# A tibble: 3,144 × 6
  GEOID state   county          pop region division
  <chr> <chr>   <chr>          <dbl> <chr>   <chr>
1 01001 Alabama Autauga County  58805 South  East South Central
2 01003 Alabama Baldwin County 231767 South  East South Central
3 01005 Alabama Barbour County  25223 South  East South Central
# i 3,141 more rows
```

Joining real data: multiple syntaxes

Current, recommended syntax:

```
counties_joined ← counties |>  
  left_join(county_pop, join_by(GEOID = fips))
```

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Older, but also valid syntax:

```
counties_joined <- counties |>  
  left_join(county_pop, by = c('GEOID' = 'fips'))
```

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  left_join(county_pop, by = c('GEOID' = 'fips'))
```

Most replication data (and many of your colleagues) will use the older syntax!

New syntax allows for *fancier* joins, like `join_by(x > x)`...

Find unmatched keys

Check the rows that do not match:

```
counties |>
  select(GEOID, state, county) |>
  anti_join(county_pop, join_by(GEOID = fips)) |>
  select(GEOID, state, county)
```

```
# A tibble: 9 × 3
  GEOID state      county
<chr> <chr>      <chr>
1 09110 Connecticut Capitol Planning Region
2 09120 Connecticut Greater Bridgeport Planning Region
3 09130 Connecticut Lower Connecticut River Valley Planning Region
# i 6 more rows
```

Find unmatched keys

Check the rows that do not match:

```
counties |>
  select(GEOID, state, county) |>
  anti_join(county_pop, join_by(GEOID = fips)) |>
  select(GEOID, state, county)
```

```
# A tibble: 9 × 3
  GEOID state      county
<chr> <chr>      <chr>
1 09110 Connecticut Capitol Planning Region
2 09120 Connecticut Greater Bridgeport Planning Region
3 09130 Connecticut Lower Connecticut River Valley Planning Region
# i 6 more rows
```

What's wrong with Connecticut?

Duplicate keys change row counts

A join can duplicate left rows when the right key is not unique

```
attitudes_duplicate <- tibble(  
  id = c(101L, 101L, 102L),  
  trust_government = c(3L, 4L, 5L)  
)  
  
party |>  
  left_join(attitudes_duplicate, join_by(id))
```

```
# A tibble: 4 × 3  
   id party      trust_government  
  <int> <chr>          <int>  
1   101 Democrat           3  
2   101 Democrat           4  
3   102 Independent        5  
# i 1 more row
```

purrr

purrr

map()

`map(x, f)` is a new way to do yesterday's `lapply(x, f)`

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Why `map()` then? Large family of *type-stable* `map_*()` functions!

And bits of friendlier syntax:

```
lapply(x, `[`, "a")  
map(x, "a")
```

Typed maps

Typed maps return a vector of a *known* type:

- `map_dbl()`: double vector

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Typed maps

Typed maps return a vector of a *known* type:

- `map_dbl()`: double vector
- `map_int()`: integer vector
- `map_chr()`: character vector
- `map_lgl()`: logical vector

```
counties |>
  select(income, rent) |>
  map_dbl(
    \(x) mean(x, na.rm = TRUE)
  )
```

income	rent
66073.0945	938.1388

Mapping over data frames

Create a list of tibbles with `group_split()`:

```
counties_by_region <- counties_joined |>  
  filter(!is.na(region)) |>  
  group_by(region) |>  
  group_split()
```

Mapping over data frames

Create a list of tibbles with `group_split()`:

```
counties_by_region <- counties_joined |>
  filter(!is.na(region)) |>
  group_by(region) |>
  group_split()
```

`map_dbl()` then returns one value per tibble:

```
map_dbl(
  counties_by_region,
  \(x) mean(x$income, na.rm = TRUE)
)
```

```
[1] 67678.39 78195.17 60859.68 72626.80
```

purrr

Combining results

First return one tibble for each region

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First return one tibble for each region

```
summaries <- map(  
  counties_by_region,  
  function(data) {  
    summarize(  
      data,  
      n = n(),  
      mean_income = mean(income, na.rm = TRUE)  
    )  
  }  
)
```

Combining results

First return one tibble for each region

```
summaries <- map(  
  counties_by_region,  
  function(data) {  
    summarize(  
      data,  
      n = n(),  
      mean_income = mean(income, na.rm = TRUE)  
    )  
  }  
)
```

```
all(map_lgl(summaries, is_tibble))  
length(summaries)
```

```
[1] TRUE  
[1] 4
```

Stack compatible results

`list_rbind()` combines the tibbles by row

```
list_rbind(summaries)
```

```
# A tibble: 4 × 2
      n mean_income
<int>      <dbl>
1  1055    67678.
2   209    78195.
3  1422    60860.
# i 1 more row
```


Several inputs

`map2( )` iterates over two (equal length) inputs in parallel

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```
district_turnout <- list(c(0.62, 0.58, 0.64), c(0.71, 0.69, 0.74))  
benchmarks <- c(0.60, 0.70)
```

```
map2_lgl(  
  district_turnout,  
  benchmarks,  
  \(turnout, benchmark) mean(turnout) > benchmark  
)
```

```
[1] TRUE TRUE
```

Applied exercise

Exercise

Using `counties_joined`, compare income by region and population level

- Label counties as above or below the median population

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Using `counties_joined`, compare income by region and population level

- Label counties as above or below the median population
- Compute the number of counties and mean income by region and population level
- Order the summary from highest to lowest mean income

Write one pipeline

Step 1: label population

```
counties_labeled ← counties_joined |>
  mutate(
    pop_level = # ...
  )
```


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```
counties_labeled ← counties_joined |>
  mutate(
    pop_level = # ...
  )
```

Use `if_else()` or `case_when()`

Step 2: summarize groups

```
result <- counties_labeled |>  
  # group by region and population level  
  # summarize n and mean_income  
  # arrange by descending mean_income
```

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```
result <- counties_labeled |>  
  # group by region and population level  
  # summarize n and mean_income  
  # arrange by descending mean_income
```

How many rows should the result contain?

One possible solution

```
result <- counties_joined |>
  filter(!is.na(pop), !is.na(region)) |>
  mutate(
    pop_level = if_else(
      pop >= median(pop),
      'at or above median',
      'below median'
    )
  ) |>
  group_by(region, pop_level) |>
  summarize(
    n = n(),
    mean_income = mean(income, na.rm = TRUE),
    .groups = 'drop'
  ) |>
  arrange(desc(mean_income))
```

Apply a function to each group

```
region_summary <- function(data) {  
  data |>  
    summarize(  
      n = n(),  
      mean_income = mean(income, na.rm = TRUE),  
      median_rent = median(rent, na.rm = TRUE)  
    )  
}
```

Apply a function to each group

```
region_summary <- function(data) {  
  data |>  
    summarize(  
      n = n(),  
      mean_income = mean(income, na.rm = TRUE),  
      median_rent = median(rent, na.rm = TRUE)  
    )  
}
```

Apply `region_summary()` to each region with `map()`

One possible solution

```
by_region <- counties_joined |>
  filter(!is.na(region)) |>
  group_by(region) |>
  group_split()
```

```
by_region |>
  map(region_summary) |>
  list_rbind()
```

```
# A tibble: 4 × 3
      n mean_income median_rent
  <int>      <dbl>      <dbl>
1  1055    67678.         804
2   209    78195.        1056
3  1422    60860.         845
# i 1 more row
```

Other tidyverse packages

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- `stringr` works with character strings
- `lubridate` parses and transforms dates and times
- `forcats` works with categorical variables (“factors”)
- `readr` imports rectangular data files

Summary

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- Start with a tibble and inspect its rows, columns, and types
- Use one dplyr verb for each transformation
- Use pipes to connect transformations
- Group before computing group-level summaries
- Check keys before and after joins
- Use `map()` to apply a function to every list element